

JSON vs Hash

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In your Redis code, **JSON vs Hash** means:

- In one API, you are storing the whole user object as a single JSON string.
- In another API, you are storing user fields separately using Redis Hash.

1. JSON Storage

```
await redis.set(  
  `user:${req.params.id}:json`,  
  JSON.stringify(req.body)  
);
```

Suppose request body is:

```
{  
  "name": "Atul",  
  "age": 21,  
  "city": "Agra"  
}
```

Then Redis stores:

Key:

user:1:json

Value:

'{"name": "Atul", "age": 21, "city": "Agra"}'

Everything is stored as **one string**.

To read it back:

```
const data = await redis.get(...);  
JSON.parse(data)
```

because Redis returns string data.

2. Hash Storage

```
await redis.hset(  
  `user:${req.params.id}:hash`,  
  req.body  
);
```

Redis stores data like:

Key:

user:1:hash

Fields:

name -> Atul

age -> 21

city -> Agra

This is called a **Redis Hash**.

Each property becomes its own field.

To get all fields:

```
await redis.hgetall(...)
```

Difference Between JSON and Hash

JSON	Hash
Stored as one string	Stored as separate fields
Uses <code>set()</code> and <code>get()</code>	Uses <code>hset()</code> and <code>hgetall()</code>
Need <code>JSON.stringify()</code>	No stringify needed
Need <code>JSON.parse()</code>	No parse needed
Updating one field is harder	Updating one field is easy
Good for complete objects	Good for field-based access

Example

JSON Update

If you want to change only age:

First:

1. get full object
2. parse
3. modify
4. stringify again

5. save again

```
const data = JSON.parse(await redis.get(key));

data.age = 22;

await redis.set(key, JSON.stringify(data));
```

Hash Update

Directly update single field:

```
await redis.hset(key, "age", 22);
```

Much easier.

When to Use What

Use JSON when:

- You always need full object
- Nested objects exist
- Structure is complex

Example:

```
{
  "name": "Atul",
  "address": {
```

```
"city": "Agra",  
"pin": 282001  
}  
}
```

Use Hash when:

- Simple flat data
- Need individual field updates
- Need better Redis-style operations

Example:

```
{  
  "name": "Atul",  
  "age": 21  
}
```

Your APIs

JSON APIs

```
POST /user/:id/json  
GET /user/:id/json
```

Store and retrieve complete object as JSON string.

Hash APIs

```
POST /user/:id/hash
```

```
GET /user/:id/hash
```

Store and retrieve data using Redis Hash fields.

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